Faishal Uddin Himel

Basundhara R/A, Block-2C
01987779101
himelfaishal@gmail.com
linkedin.com/in/faishal-uddin-himel-b7b29a236/
https://my-portfolio-cyan-nu-69.vercel.app/
GitHub

CAREER OBJECTIVE

I'm passionate about building intelligent, scalable backend systems that turn AI ideas into real, production-ready products. With a strong foundation in AI and backend development, I focus on designing secure APIs, efficient data pipelines, and reliable architectures that solve complex problems. I'm committed to continuous learning, hands-on experimentation, and delivering high-impact, future-ready solutions through clean engineering and practical innovation.

PROFILE SUMMARY

- Highly motivated and research-driven AI & Backend Development student with a strong foundation in scalable backend systems, intelligent automation, and modern software engineering. Demonstrates strong academic performance with a consistent record of excellence.
- Experienced in building and integrating AI-driven features into backend platforms, including model-serving APIs, data processing pipelines, authentication/authorization, and secure service architectures for real-world problem-solving and research-focused development.
- Proficient in Python, JavaScript/TypeScript, SQL, C++, C#, Java, PHP, HTML, CSS, and R. Skilled in developing AI/ML models and deploying them via secure RESTful APIs using FastAPI and Node.js/Express, with hands-on experience in Next.js backend features (API routes/server actions), PostgreSQL, and ORM-based database design.
- Comfortable with production workflows and tooling including Git, Docker, Linux, CI/CD basics, and cloud-ready deployment practices. Experienced with TensorFlow, PyTorch, and Scikit-learn for AI development and backend integration.
- Passionate about developing intelligent, secure, and reliable backend solutions through collaborative teamwork. Recognized for leadership, communication skills, continuous learning, and delivering impactful, scalable systems.

WORK EXPERIENCE

AI Developer & Automation Engineer
Betopia Group, Bangladesh [On-site/Remote] January 2026 – Present

- Developing and deploying AI-powered features for internal products and client projects.
- Designing automation workflows to streamline business processes and reduce manual operations.
- Building backend services and integrations (APIs, databases, third-party tools) to support AI/automation pipelines.
- Implementing data processing, model evaluation, and performance improvements for production readiness.
- Collaborating with cross-functional teams to deliver end-to-end solutions and maintain documentation.

Research Intern
Applied Intelligence and Informatics Lab (AIIL), Nottingham, UK [Remote] October 2025 – December 2025

- Conducted research aligned with AIIL's core focus areas in applied artificial intelligence and informatics.
- Performed in-depth literature reviews to support model development and experimental design.
- Designed, implemented, and evaluated machine learning and AI research models.
- Collaborated with international researchers through virtual meetings and technical discussions.
- Contributed to research documentation and journal publication preparation targeting Q1/Q2 indexed journals.

Leader
Team Tech Wing, American International University-Bangladesh (AIUB) October 2023 – Present

- Supervised all laboratory activities to ensure seamless operations and effective utilization of technical resources.
- Provided mentorship and guidance to junior members, assisting in their technical and professional skill development.
- Organized and led knowledge-sharing sessions and technical workshops to promote collaborative learning within the team.

EDGE Project
AIUB Institute (EDGE Project – Enhancing Digital Government and Economy) June 2024 – January 2025

- Worked on a project-based assignment using Python, contributing to research and development under the national EDGE initiative.
- Collaborated with academic supervisors to implement digital solutions aligned with project goals.
- Supervisor: Dr. Mohammad Saef Ullah Miah (saef@aiub.edu)

Intern App Developer
RIC-Series August 25, 2025 – November 10, 2025

- Assisting in the development of gaming applications and contributing to codebase enhancements.
- Collaborating with development teams to improve application performance and user experience.

WORK INTERESTS

- AI Agents & Multi-Agent Systems
- Backend Development (Node.js / Next.js / FastAPI)
- RESTful APIs & System Integration
- AI Automation & Workflow Orchestration
- Data Engineering & Analytics
- MLOps & Model Deployment

RESEARCH PROJECTS

AI-Powered Endpoint Detection and Response (EDR) System with API Integration

- Supervisor: Dr. Mohammad Saef Ullah Miah
- Developed an intelligent endpoint protection system using machine learning for real-time malware and anomaly detection.

- Designed and implemented secure RESTful APIs to enable communication between endpoints and the central detection engine.
- Integrated automated threat response mechanisms, including process isolation and alert generation.
- Applied feature extraction and model optimisation techniques to improve detection accuracy and reduce false positives.
- Evaluated system performance using real-world malware and benign datasets.

Multi-Agent AI CyberShield with Three Cooperative Detection Agents

- Supervisor: Dr. Mohammad Saef Ullah Miah
- Designed a distributed CyberShield architecture using three cooperative AI-based detector agents for network, host, and application-level threat detection.
- Implemented independent machine learning models within each agent to detect malware, intrusion, and anomalous behavior in real time.
- Developed an agent coordination and decision-fusion mechanism to improve detection accuracy and reduce false alarms.
- Integrated the system with a central response unit for automated alerting and threat mitigation.
- Validated the multi-agent framework using mixed benign and attack datasets in a simulated enterprise–IoT environment.

Multimodal AI System for Rice Disease Detection with Environmental Context

- Supervisor: Dr. Afroza Nahar
- Designed a multimodal crop intelligence framework integrating leaf image analysis with environmental data for rice disease detection.
- Implemented an EfficientNet-based deep learning model to extract visual disease features from rice leaf images.
- Developed a tabular environmental encoder to process weather and soil variables including temperature, humidity, rainfall, soil moisture, and nutrient indices.
- Proposed a feature-level fusion architecture combining visual and environmental embeddings to improve disease classification performance.
- Applied explainability techniques including Grad-CAM and permutation feature importance to interpret model decisions.
- Constructed a context-enriched dataset of 30,834 rice leaf images augmented with satellite-derived meteorological data and agronomic soil indicators.

Adversarial Learning–Based Phishing Email Detection

- Supervisor: Mir Md. Kawsur
- Developed a phishing email detection framework using NLP and adversarial learning techniques.
- Implemented a transformer-based classifier (DistilBERT) trained on real and adversarially modified email samples.
- Applied adversarial example generation (e.g., TextFooler, synonym substitution, URL obfuscation) to improve robustness against phishing evasion.
- Evaluated the system using phishing and benign email datasets to analyze detection performance.

PROJECTS

LLM-Driven Reflective Multi-Agent Economic Simulation System 2024 – 2025
Microsoft Imagine Cup *American International University-Bangladesh*

- Built a multi-agent economic simulation modeling heterogeneous individuals under real-world constraints such as wages, education, consumption, and savings.

- Designed a hybrid decision architecture where rule-based economics enforces feasibility and LLMs act as a reflective policy layer for adaptive labor decisions.
- Implemented LLM-based reflection memory enabling agents to reason over past outcomes while preserving mathematical and budget constraints.
- Integrated rate-limited LLM invocation and simulated economic shock scenarios to evaluate resilience and behavioral adaptation.
- Produced interpretable analytics and visualizations (labor trends, savings by education, mood dynamics) to validate emergent behaviour.
- **Technologies:** Python, OpenAI GPT-4o / GPT-4o-mini, Multi-Agent Systems, Prompt Engineering, NumPy, Pandas, Matplotlib

CyberShield: Reflective Multi-Agent Zero-Day Cyber Defense System 2025
Independent Research / Academic Project *American International University-Bangladesh*

- Designed an end-to-end autonomous cyber defense system for zero-day attack detection using unsupervised GAN-based anomaly detection.
- Implemented a WGAN-GP critic trained exclusively on normal network traffic to identify previously unseen attacks without labeled data.
- Developed a Coordinator Agent to perform temporal correlation of anomalies, escalating isolated anomalies into confirmed attack campaigns.
- Built a Responder Agent capable of automated defense actions, including monitoring, host isolation, and IP blocking based on attack severity.
- Integrated reflective incident memory and feedback loops to reduce false positives and enable campaign-level reasoning.
- Produced comprehensive visual analytics, including anomaly score distributions, temporal escalation timelines, attack lifecycle diagrams, and explainability overlays.
- **Technologies:** Python, TensorFlow, WGAN-GP, GAN-Based Anomaly Detection, Multi-Agent Systems, Cybersecurity Analytics, NumPy, Pandas, Matplotlib

SENTRI: Smart Edge-Enabled Threat Response & Intelligence System 2025
Independent Research / Academic Project *American International University-Bangladesh*

- Designed an edge-enabled threat detection and response system for IoT/endpoint environments, optimized for low-latency monitoring and lightweight deployment.
- Built a real-time anomaly/intrusion detection pipeline using network-flow and device telemetry features with streaming window-based inference.
- Implemented an Alert Prioritization module to score severity, suppress noisy alerts, and generate actionable incident summaries.
- Developed an Automated Response component supporting policy-driven actions such as IP/port blocking, suspicious process quarantine, traffic throttling, and device isolation triggers.
- Integrated incident memory and feedback-based threshold tuning to reduce recurring false positives and improve stability under evolving traffic patterns.
- Produced security visualizations including anomaly score trends, event timelines, attack-path/response flow diagrams, and model performance dashboards.
- **Technologies:** Python, Edge AI, Network Security, IoT Security, Scikit-learn, TensorFlow/PyTorch, NumPy, Pandas, Matplotlib

Student Workspace: AI + Automation University Management & Student Productivity Platform 2026
SaaS / MVP Project *Independent Project*

- Designed a unified student + university operations platform combining smart semester registration, syllabus-driven study planning, live class management, and academic integrity support.

- Built a course registration optimizer to validate prerequisites, detect timetable conflicts, check seat availability, and recommend best-fit sections for 4–5 courses with alternatives when sections are full.
- Implemented waitlist automation to store unmet registration requests and automatically notify students when seats open or new sections are created, reducing manual advising workload.
- Developed a syllabus ingestion pipeline (PDF/text) to extract weekly topics, grading breakdowns, and key dates, converting unstructured content into a normalized timeline with daily checklists and progress tracking.
- Implemented an AI planning layer to generate weekly study schedules, 7/14/30-day revision plans, and an Exam Readiness Score with behind-schedule alerts and recovery recommendations.
- Integrated a lightweight local LLM assistant (Ollama) for intent → constraints, decision explanations, and notes-to-tasks assistance while keeping eligibility/seat logic deterministic.
- Added academic integrity & writing support using similarity-risk analysis (overlap + embeddings) with evidence highlighting, citation guidance, and grammar/clarity improvements for student learning.
- Built role-based dashboards (student/teacher/admin) with analytics, exports, and audit logs; implemented event-driven notifications (webhooks + background workers) for confirmations, reminders, and expiry alerts.
- **Technologies:** Next.js, TypeScript, TailwindCSS, FastAPI, PostgreSQL, Prisma/SQLAlchemy, Redis, Celery (Workers/Scheduler), Ollama (Local LLM), Embeddings Similarity, PDF/Text Parsing, REST APIs, Webhooks, Git

AI-Enhanced Endpoint Malware Protection with Adversarial Robustness 2025
Research Internship Project *Applied Intelligence and Informatics Lab (AIIL), UK*

- Designed and implemented an end-to-end AI-based endpoint malware detection pipeline using static feature analysis of executable files.
- Developed transformer-based deep learning models alongside classical machine learning baselines for robust malware classification.
- Integrated adversarial robustness mechanisms, including GAN-based malware augmentation and PGD adversarial attacks, to evaluate model resilience against evasion.
- Performed extensive experimental evaluation using ROC-AUC, PR-AUC, precision-recall, calibration analysis, and adversarial distance metrics.
- Analyzed the impact of synthetic data augmentation under different filtering thresholds to balance generalization and overfitting.
- Visualized malware and synthetic sample distributions using t-SNE and feature-space analysis to validate distributional fidelity.
- Exported trained models to ONNX format and verified consistency between PyTorch and deployment-ready inference.
- **Technologies:** Python, PyTorch, Transformer Models, GANs, PGD Attacks, ONNX, Scikit-learn, NumPy, Pandas, Matplotlib

EDGE Project – Enhancing Digital Government and Economy June 2024 – January 2025

- Contributed to national-level digital transformation efforts through Python-based system development.
- Supervisor: Dr. Mohammad Saef Ullah Miah (saef@aiub.edu)

PERSONAL QUALIFICATIONS

- Strong analytical and problem-solving ability with experience in designing AI-driven and backend systems.
- Effective communicator with experience collaborating with international researchers and cross-functional development teams.

- Fast learner with the ability to quickly adapt to new technologies, frameworks, and research methodologies.
- Proven leadership and mentoring experience through supervising technical teams and guiding junior members.
- Capable of managing multiple responsibilities and delivering high-quality results under tight deadlines.
- Self-motivated, disciplined, and committed to continuous learning, research, and innovation.

CERTIFICATES & TRAINING

LLM Pentesting: Mastering Security Testing for AI Models
Udemy – Certificate of Completion *June 2024*

Artificial Intelligence & Machine Learning Fundamentals
Certificate of Completion *May 2024*

Generative AI: Prompt Engineering Basics
IBM – Certificate of Completion *April 2024*

Professional Web, Android & iOS Penetration Testing
Cyber-Bangla – Certificate of Completion *October 2024*

Python for Data Science and Machine Learning
IBM – Certificate of Completion *September 2024*

Python Django Development Course
IIT, Jahangirnagar University *December 2024*

Advanced Cybersecurity Course
Team Matrix – Elite Hackers *March 2025*

Cisco Certified Network Associate (CCNA)
AIUB Institute of Continuing Education
Cisco *August 2025*

EDUCATION

Bachelor of Science in Computer Science and Engineering
American International University-Bangladesh (AIUB) 2022 – 2025 (Completed)

Higher Secondary Certificate (HSC)
Pakundia Govt. College 2018 – 2020

- Major: Science

REFERENCES

Prof. Dr. Dip Nandi
Associate Dean
Faculty of Science and Technology
American International University-Bangladesh
Email: `dip.nandi@aiub.edu`

Dr. Md. Saef Ullah Miah
Associate Professor, Additional Director (IQAC)
Department of Computer Science
American International University-Bangladesh
Email: `saef@aiub.edu`